

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 16

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Consider the following C code segment:

```
int a = 10;
int b = 20;

#pragma omp parallel private(a)
{
    int c = 30;
    a = omp_get_thread_num();
    b = b + a;
}
```

- Which variables are shared across all threads?
- Which variables are private to each thread?
- If this code is run with 4 threads, what is the value of b at the end? Explain why this results in a race condition.

2. Write a short C code snippet using OpenMP that does the following:

- Sets the number of threads to 8 using a library function.
- Uses a parallel directive to have each thread print: "Hello from thread [ID] out of [Total] threads." Hint: You will need to use `omp_get_thread_num()`, `omp_get_num_threads()`, and `omp_get_num_procs()`.

• Welcome: Sean

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